

11	(a)	(i)	metal (<i>allow specific example of a metal</i>)	B1	[1]
		(ii)	e.g. provides 'return' for the signal shields inner core from interference/reduces cross-talk/reduces noise increased security (<i>any two sensible suggestions, 1 each</i>)	B2	[2]
	(b)	(i)	(gradual) loss of power/intensity/amplitude	B1	[1]
		(ii)	dB is a log scale <i>either</i> large (range of) numbers are easier to handle (on a log scale) <i>or</i> compounding attenuations/amplifications is easier	B1 B1	 [2]
	(c)		attenuation = $190 \times 11 \times 10^{-3} = 2.09 \text{ dB}$ $-2.09 = 10 \lg(P_{\text{OUT}}/P_{\text{IN}})$ ratio = 0.62	C1 C1 A1	 [3]

Page 6	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2015	9702	42